

Question	Answer	Marks
7(a)(i)	$T = 2\pi / 40\pi = 0.050 \text{ s}$	A1
7(a)(ii)	$V_{\text{r.m.s.}} = 18 / \sqrt{2}$ $= 13 \text{ V}$	A1
7(b)	sinusoidal curve of period 50 ms from $t = 0$ to $t = 100 \text{ ms}$	B1
	correct phase (V_{MAX} at $t = 0, 50, 100 \text{ ms}$ and $-V_{\text{MAX}}$ at 25, 75 ms etc.)	B1
	maximum and minimum voltages shown as $\pm 18 \text{ V}$	B1
7(c)	<i>Any three points from:</i> - rectification is full-wave - mean power = 14 W - resistance of $R = 12 \Omega$ - peak current in $R = 1.6 \text{ A}$ or r.m.s. current in $R = 1.1 \text{ A}$ - period of <u>output</u> voltage / power = 25 ms or frequency of <u>output</u> voltage / power = 40 Hz or angular frequency of <u>output</u> voltage / power = 250 rad s^{-1}	B3

9702/42

Cambridge International AS & A Level – Mark Scheme

October/November 2025

PUBLISHED

Question	Answer	Marks